

AI-Powered Data Insights Dashboard

Objective

This project is a simple web dashboard where you upload any CSV file and it automatically tells you what's going on inside your data no coding or spreadsheet skills needed. It shows you the shape of your data, points out missing or duplicate entries, draws charts so you can see patterns visually, and then uses AI to write up a plain-English summary with practical recommendations.

The idea was to build something a non-technical person say, a manager looking at sales numbers could open, upload a file into, and actually understand without needing anyone to explain it to them.

Working:

You open the dashboard and upload a CSV file. From there, everything happens automatically, one section at a time:

1. **Dataset Overview** shows how many rows and columns you have, how many duplicate rows exist, and how many values are missing, along with a preview of the actual data.
2. **Missing Value Check** a clear table showing exactly which columns have gaps and what percentage of each column is missing.
3. **Basic Statistics** the usual summary numbers (averages, min/max, etc.) for every column.
4. **Charts** a histogram so you can see how one column's values are spread out, and a scatter plot so you can compare any two columns against each other. Underneath each chart, a plain-language caption explains what the chart is actually showing for example, "as Experience goes up, Salary tends to go up too" instead of just showing numbers.
5. **Correlation Analysis** a heatmap showing which columns move together, again explained in plain words rather than just a grid of numbers.
6. **Key Insights** —a quick automatic list of the most useful facts about the numeric columns.
7. **AI-Generated Insights** with one click, Google's Gemini AI reads a summary of your dataset and writes out a proper analysis: overall summary, key findings, trends, data quality concerns, and recommendations.

What I Used to Build It:

Python, Streamlit (for the web interface), Pandas (for the data handling), Matplotlib and Seaborn (for the charts), and Google's Gemini API (for the AI-written insights).

Improvement :

The biggest improvement over a typical version of this kind of dashboard is that every chart explains itself in normal, everyday language. Most dashboards just show you a graph and expect you to interpret it this one tells you what the graph means, in a sentence a non technical person can read and immediately understand.

Challenges :

- Making sure the AI insights feature doesn't crash the whole app if the API key is missing or something goes wrong it now shows a friendly message instead.
- I initially had a bug where picking the same column for both chart axes caused a crash; that's now handled with a clear message asking the user to pick two different columns.
- Getting the automatic chart explanations to sound natural instead of textbook statistics took a few passes of rewriting.

Conclusion :

This dashboard transforms raw CSV data into clear, meaningful insights that anyone can understand. By combining interactive visualizations, easy-to-read statistics, and AI-powered analysis, it helps users quickly understand their data and discover important patterns—without requiring any prior knowledge of data analysis.